Design of HIV-1 integrase dimer inhibitor

the subject of study Protein-Protein Interaction (PPI) between **HIV-1 integrase (CCD dimer)** and cofactor **LEDGF/p75**. The goal was to design protein binder, which blocks the allosteric pocket by mimicking BI-224436 allosteric inhibitor.

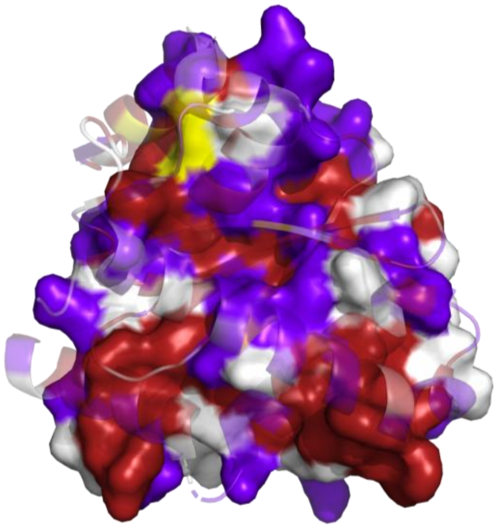
was removed water molecules, ions, and extraneous ligand, etc from all crystallographic structures.

MODULE 1: Baseline Evaluation & Small Molecule Blockade

Part I: Baseline PPI Evaluation (HIV-1 Integrase Dimer)

1.1. Identification of docking parameters and active residues

- **PRODIGY:** Delta G = -12.3 kcal/mol (Kd = 1.0). extremely strong association of monomers.
- **HADDOCK:** Stabilization occurs due to powerful electrostatics (-549.4 kcal/mol) and Van der Waals forces (-98.2 kcal/mol).
- **Receptor (Chain A) & Partner (Chain B):** Key active residues **Trp108, Glu198, Thr174, His171, Arg107.**
- **Grid Center:** X=-5.147, Y=-3.543, Z=-21.607.



1.2. Surface visualization and analysis

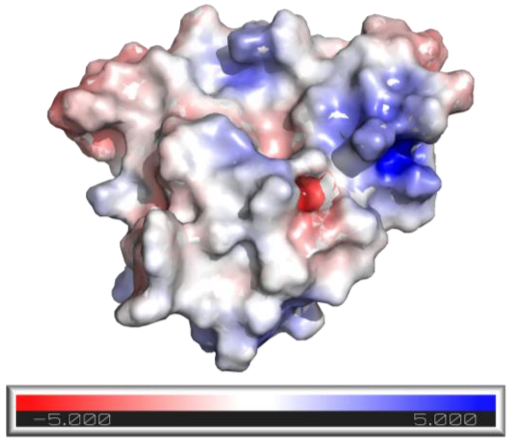
Surface properties analyses allowed to identify stability nature of HIV-1 integrase dimer.

Fig.1: Surface polarity

distribution of polar and non-polar residues on the dimer interface shows that core is mostly hydrophobic, which causes sticking of two integrase monomers.

Fig.2: Electrostatics surface

Electrostatic potential map shows high electrostatic energy due to the complementarity of the charges.



1.3. Key inter-chain contacts

Most energetically significant interactions that hold dimer together were defined.

Table 1. Critical contacts of the dimer (Top 3)

Res Chain A	Res Chain B	Bond type	Role
Arg 107	**Glu 87**	Salt-bridge	Periphery stabilization
Trp 108	**Thr 174**	H-bond / π-stacking	Core interface
His 171	**Gln 95**	H-bond	Lock the rotation angle

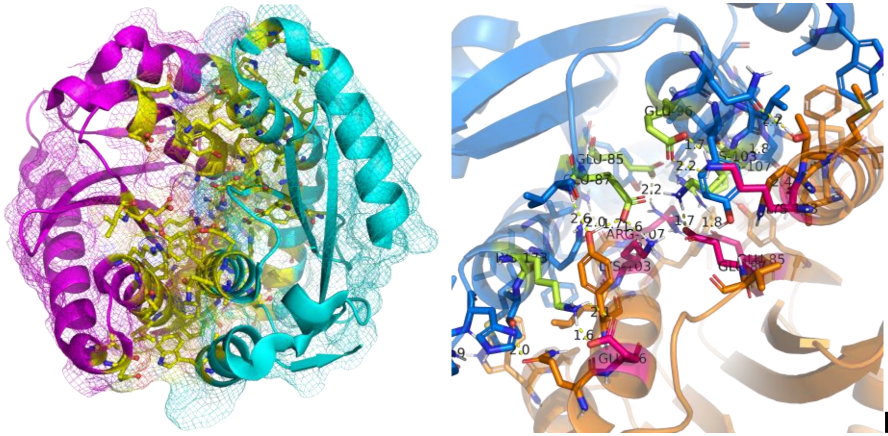
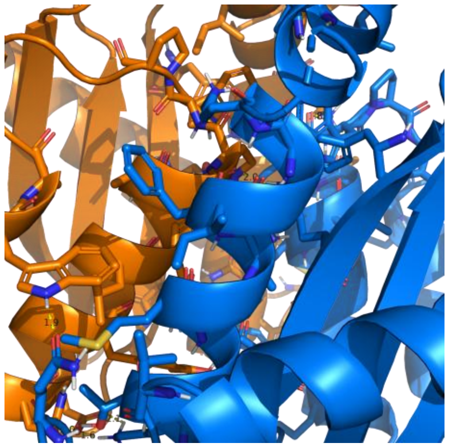


Fig.3: Contacts detalization

1.4. Dimer's final position



Best model demonstrate ideal placement of monomers.

Fig.4: Best dimer model

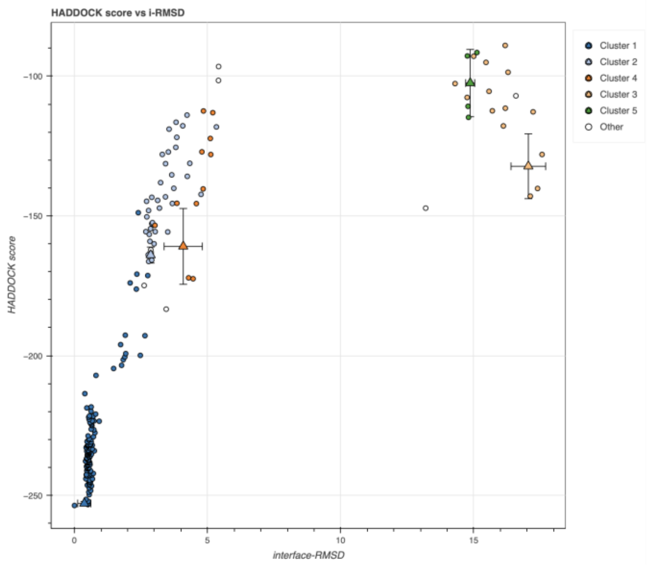
Integrase dimer is extremely stabile structure (ΔG=−12.3kcal/mol) with high contact area (~1558 Å²).

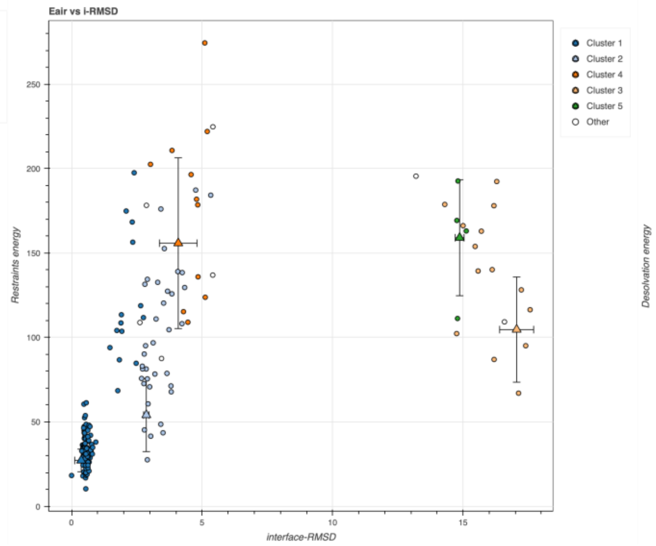
1.5. Structural Validation (Overlay with 2B4J)

to confirm docking best pose, cluster1 was aligned to experimental crystal structure 2b4j. Alignment demonstrated high accuracy RMSD = 0.626Å, confirming that HADDOCK successfully reconstructed homodimer interface without significant deviations.

1.6. HADDOCK Clustering and Energy Landscape Analysis

- **Cluster Statistics:** Cluster 1 represents most reliable prediction, with lowest average HADDOCK score and most favorable electrostatic energy (-549.4 kcal/mol).
- **Energy Funnel Concept:** i-RMSD vs HADDOCK score plots reveals a classic energy funnel. lowest energy scores tightly converge at lowest RMSD values, so prediction is not a random artifact, but thermodynamically driven.


HADDOCK score vs i-RMSD
HADDOCK score
interface-RMSD
Cluster 1
Cluster 2
Cluster 4
Cluster 3
Cluster 5
Other


Eair vs i-RMSD
Restraints energy
interface-RMSD
Cluster 1
Cluster 2
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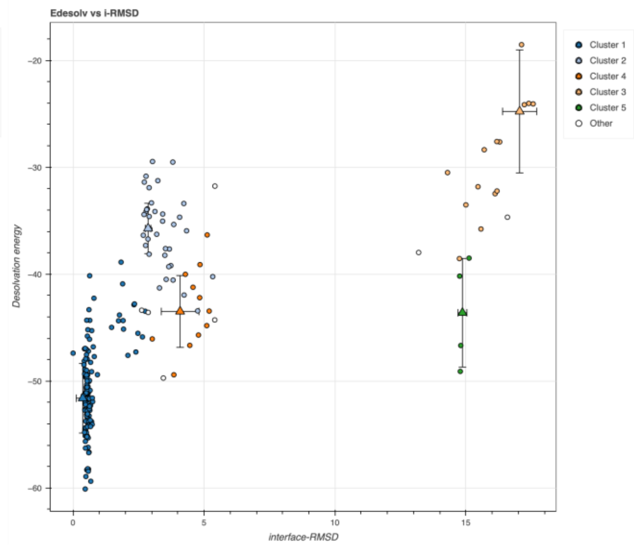
Other


Edesolv vs i-RMSD
Desolvation energy
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Cluster 1
Cluster 2
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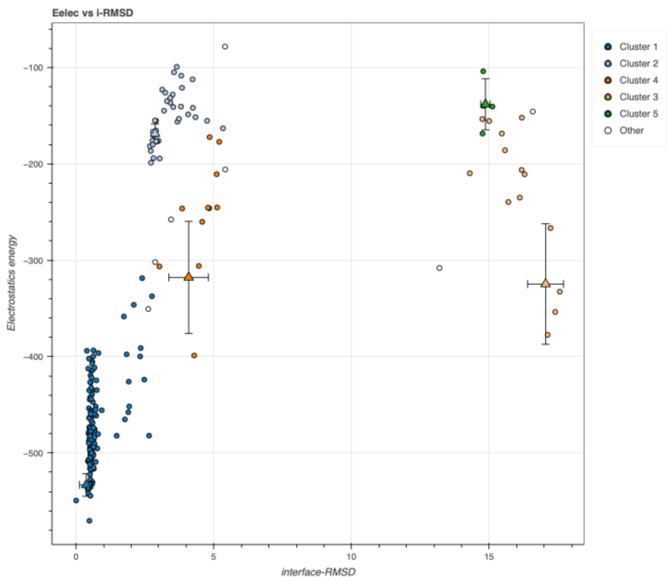
Other


Eelec vs i-RMSD
Electrostatics energy
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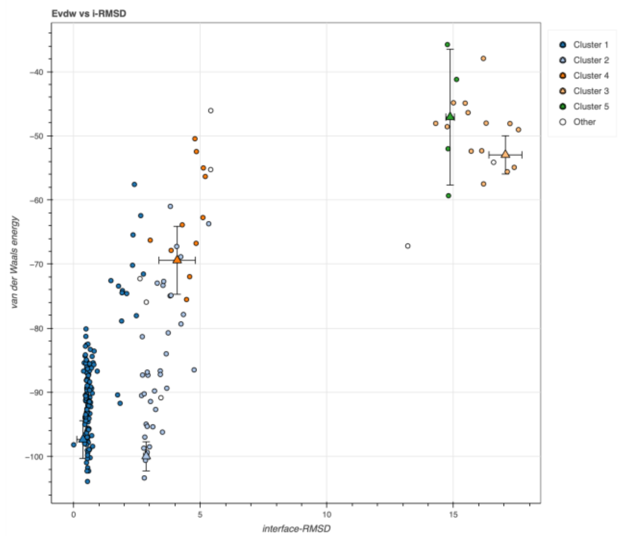
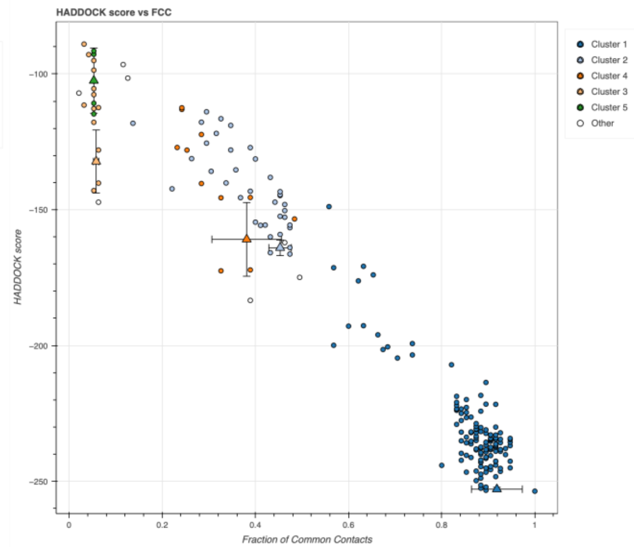
Other


Evdw vs i-RMSD
van der Waals energy
interface-RMSD
Cluster 1
Cluster 2
Cluster 4
Cluster 3
Cluster 5
Other


HADDOCK score vs FCC
HADDOCK score
Fraction of Common Contacts
Cluster 1
Cluster 2
Cluster 4
Cluster 3
Cluster 5
Other

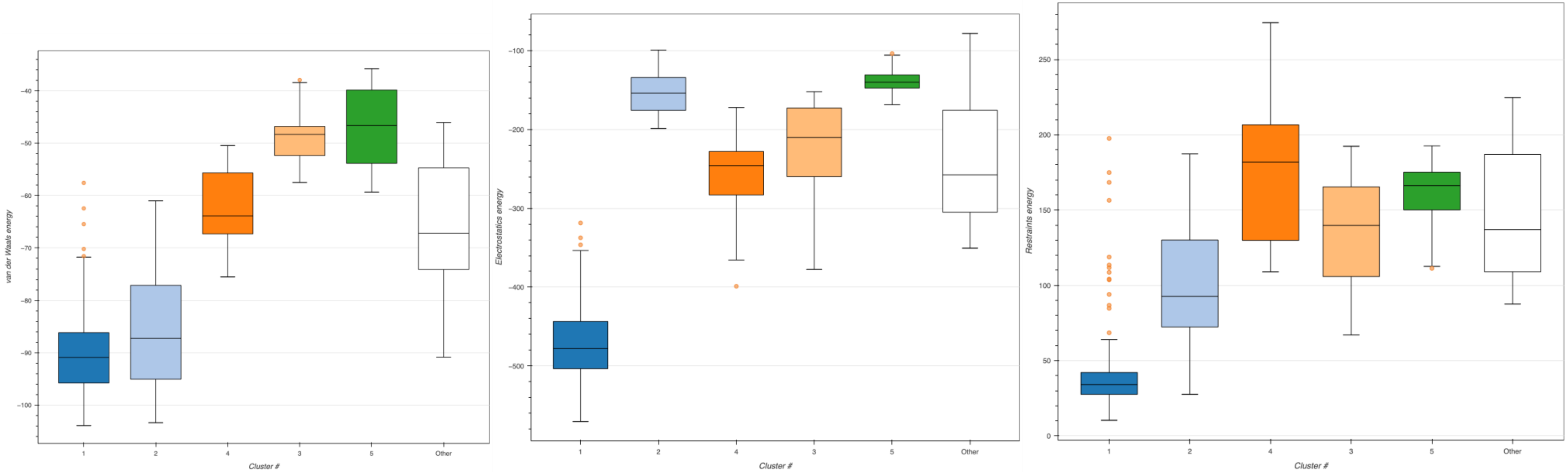
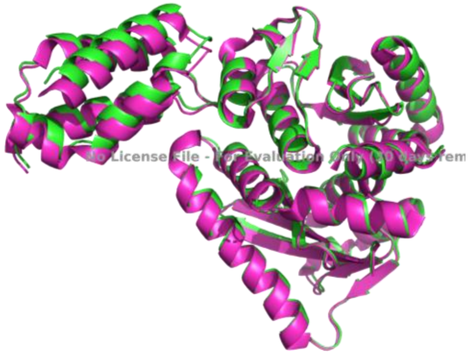
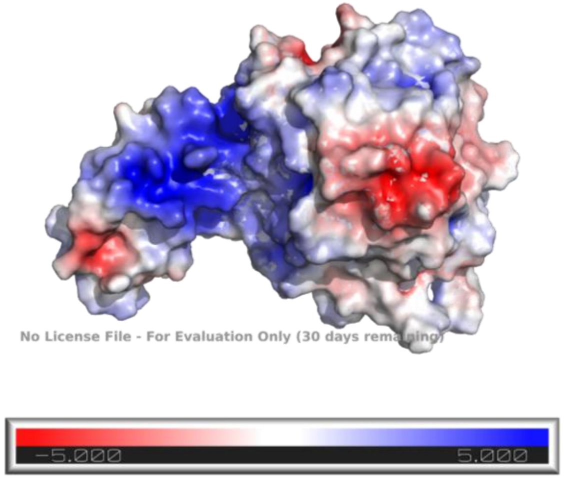


Fig.6: HADDOCK Energy and RMSD distributions



Part II: Basic PPI complex (Integrase + LEDGF)



- **PDBePISA:** interface square 628–713Å². solvation energy (-6.3 kcal/mol) confirms benefits of displacing water from pocket
- **HADDOCK:** best cluster had RMSD = 0.631Å compare to 2B4J.
- **PRODIGY:** Delta G = -8.3 kcal/mol (K_d = 0.77)

2.1. Structural validation (Overlay with 2B4J)

To validate docking results, there was structural alignment provided. RMSD = 0.631Å.

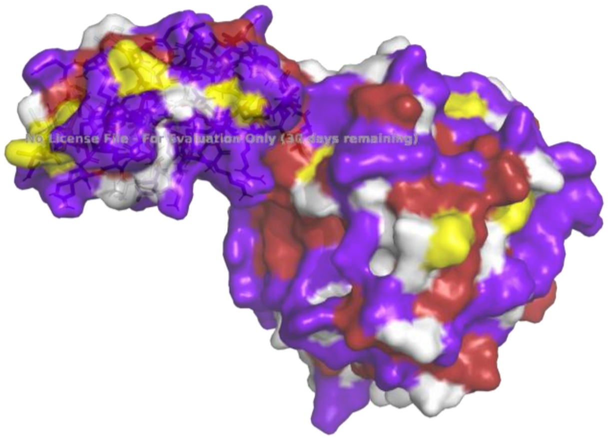


Fig.8: Overlay. Green -- cluster15, magenta -- 2BJ4.

2.2. Surface visualization and analysis

visualization clearly separates non=polar and polar regions, which allows to identificate regions where H-bonds form, and ensures corect orientation of cofactor.

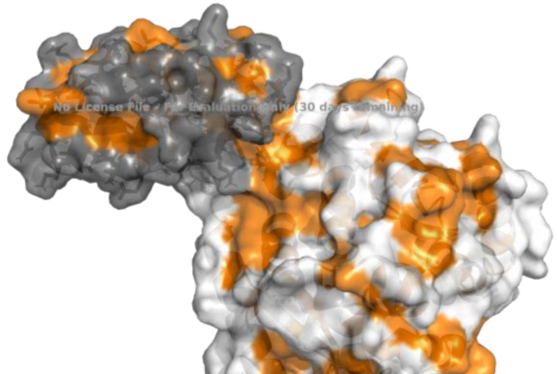


Fig.9: Surface polarity **Fig.10: Electrostatics surface**

Electrostatic surface illustrate fully positively charged LEDGF loops in binding zone.

Fig.11: Hydrophobicity surface

clearly distinguish that cofactor binds to the integrase dimer (white) precisely at the locations of distinct hydrophobic regions (orange areas)

2.3. Key H-bonds & Salt Bridges

behind hydrophobic stabilization lies network of specific polar interactions

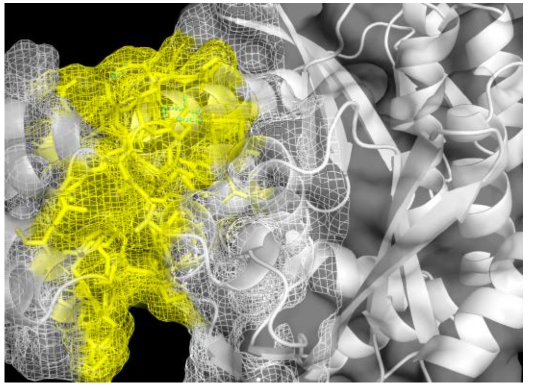


Fig.12: Contact area

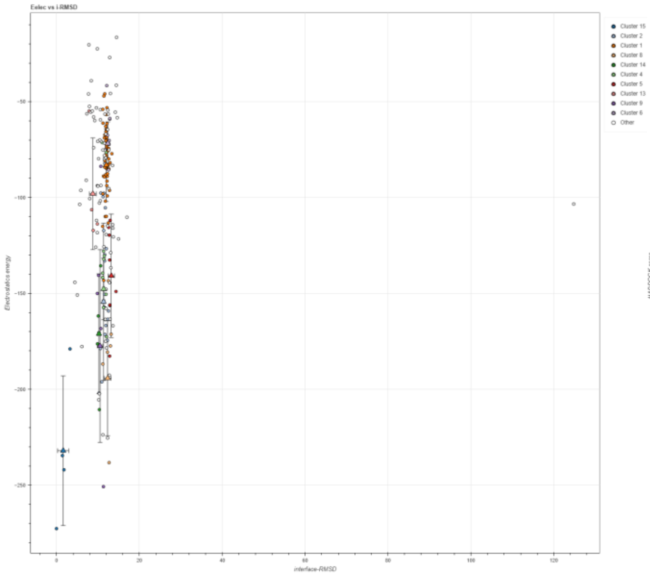
Table 2. Critical contacts of the complex (Top 3)

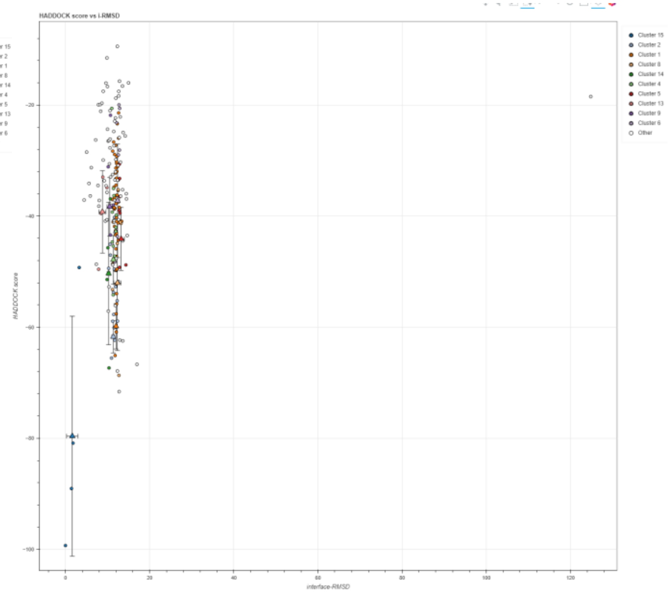
LEDGF	Integrase	Bond type	Distance (Å)
ASP 366 (O)	**LYS 173 (N)**	**Salt bridge**	**2.5 - 2.8**
THR 125 (O)	**GLN 168 (N)**	**H-bond**	**2.9**
GLU 470 (O)	**LYS 364 (N)**	**H-bond**	**3.1**

geometric fit between cofactor and dimer active sites. mesh clearly indicate that there is no free space in contact area.

2.4. HADDOCK Clustering and Energy Landscape Analysis

- **Cluster Statistics:** Cluster 15 represents most reliable prediction, with lowest average HADDOCK score and most favorable electrostatic energy (-232.05 kcal/mol).


Eelec vs i-RMSD
Electrostatic energy
interface-RMSD
Cluster 15
Cluster 2
Cluster 1
Cluster 8
Cluster 14
Cluster 4
Cluster 5
Cluster 13
Cluster 9
Cluster 6
Other


HADDOCK score vs i-RMSD
HADDOCK score
interface-RMSD
Cluster 15
Cluster 2
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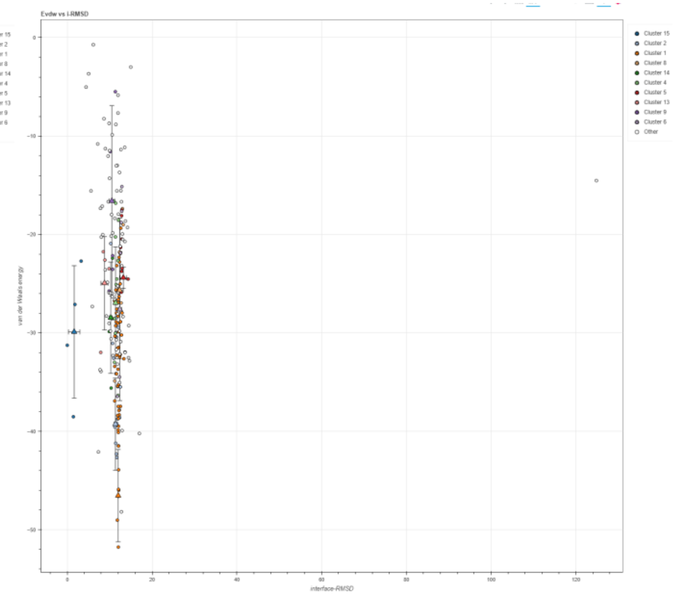
Cluster 9
Cluster 6
Other


Evdw vs i-RMSD
van der Waals energy
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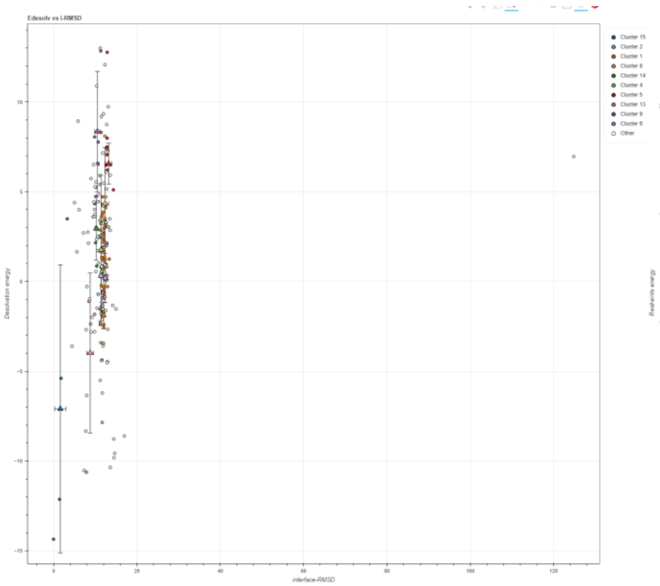
Edesolv vs i-RMSD
Desolvation energy
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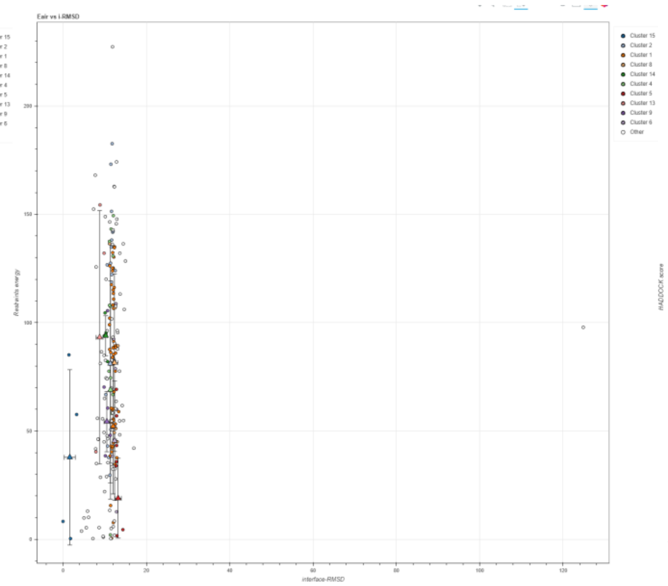
Eair vs i-RMSD
Restraints energy
interface-RMSD
Cluster 15
Cluster 2
Cluster 1
Cluster 8
Cluster 14
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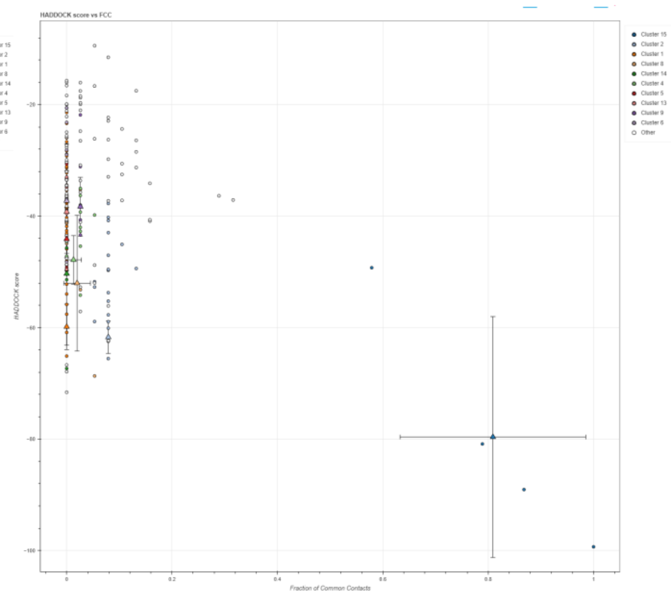
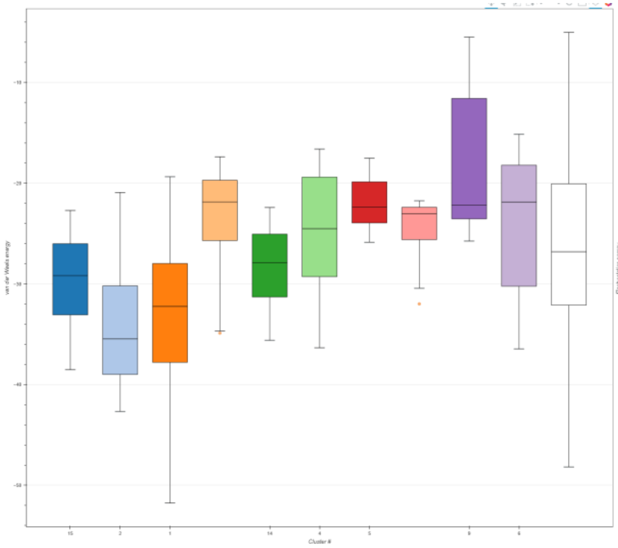
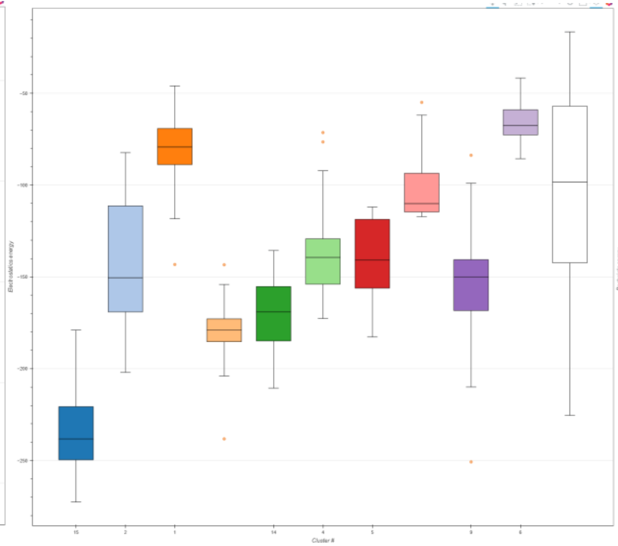
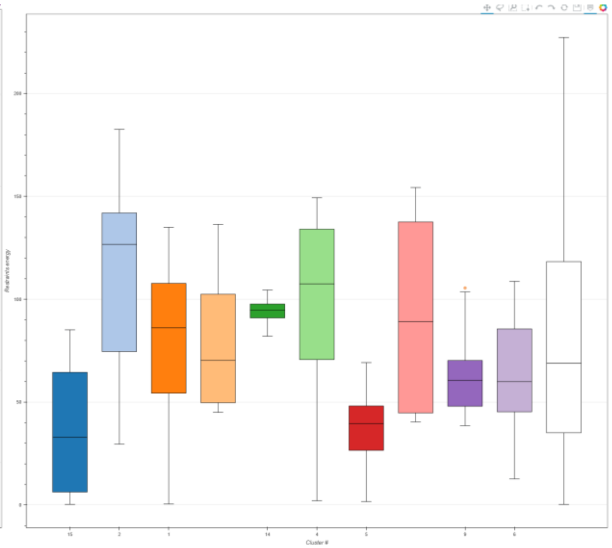
Cluster 5
Cluster 13
Cluster 9
Cluster 6
Other


HADDOCK score vs FCC
HADDOCK score
Fraction of Common Contacts
Cluster 15
Cluster 2
Cluster 1
Cluster 8
Cluster 14
Cluster 4
Cluster 5
Cluster 13
Cluster 9
Cluster 6
Other







Part III: Analysis of allosteric inhibitor (BI-224436)

3.1. HADDOCK Results

docking results of the inhibitor into allosteric pocket demonstrates high convergence and thermodynamic favorability. Cluster 1 was identified as the best

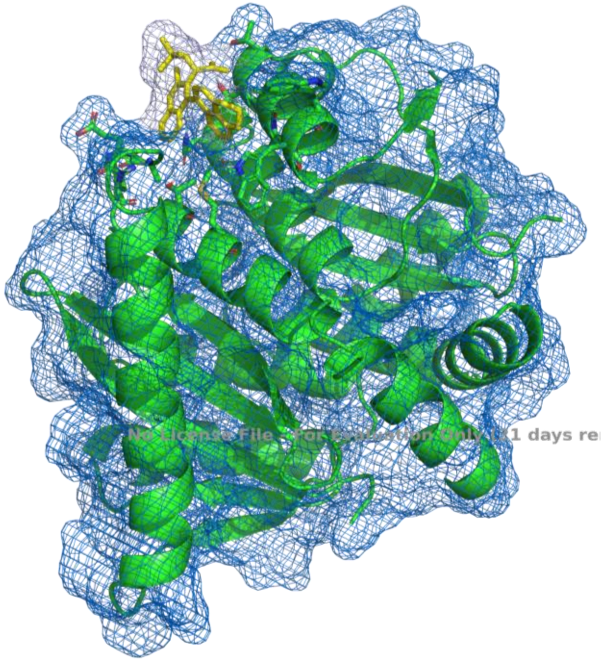


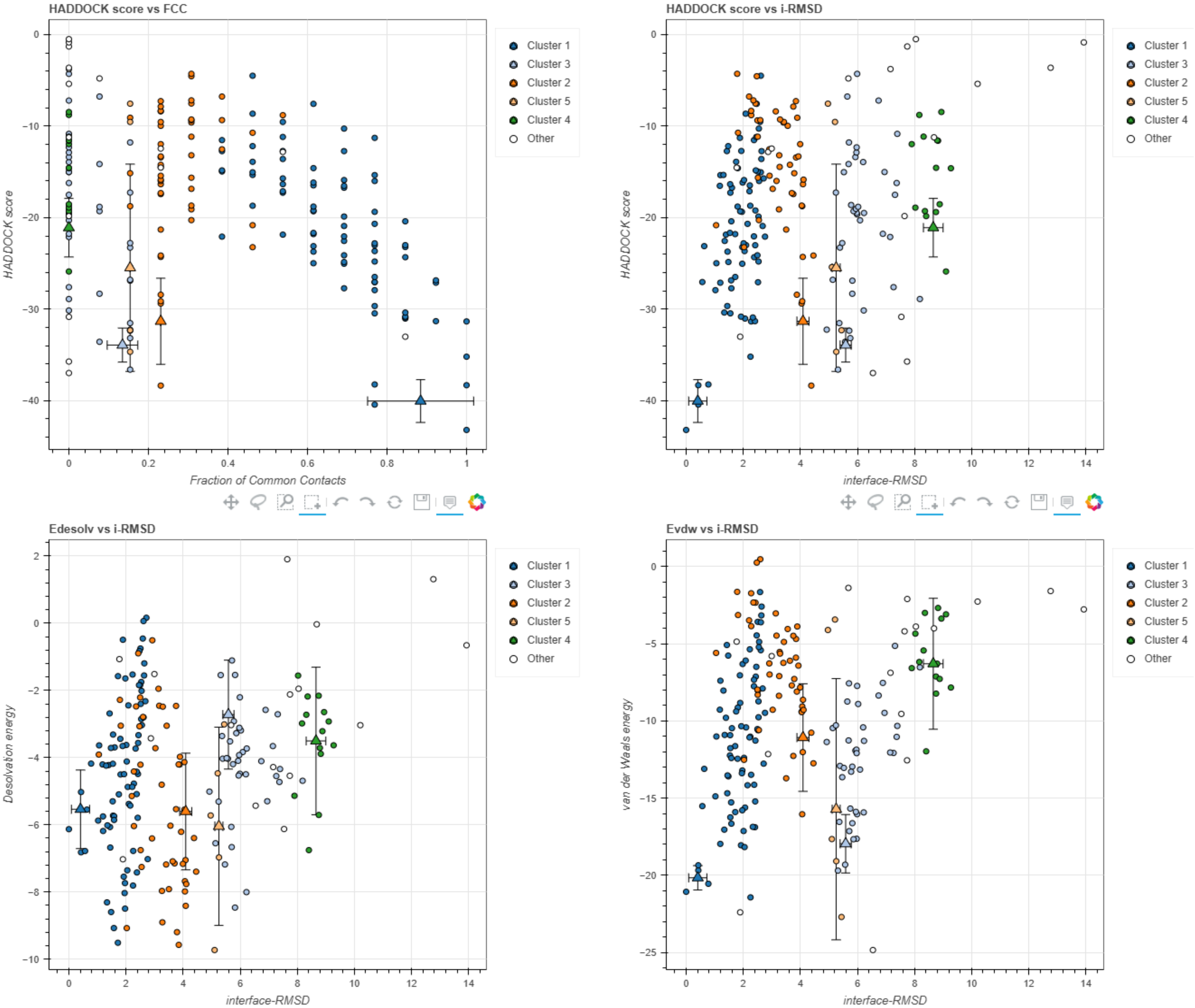
Table 3: Cluster 1 Energy characteristics

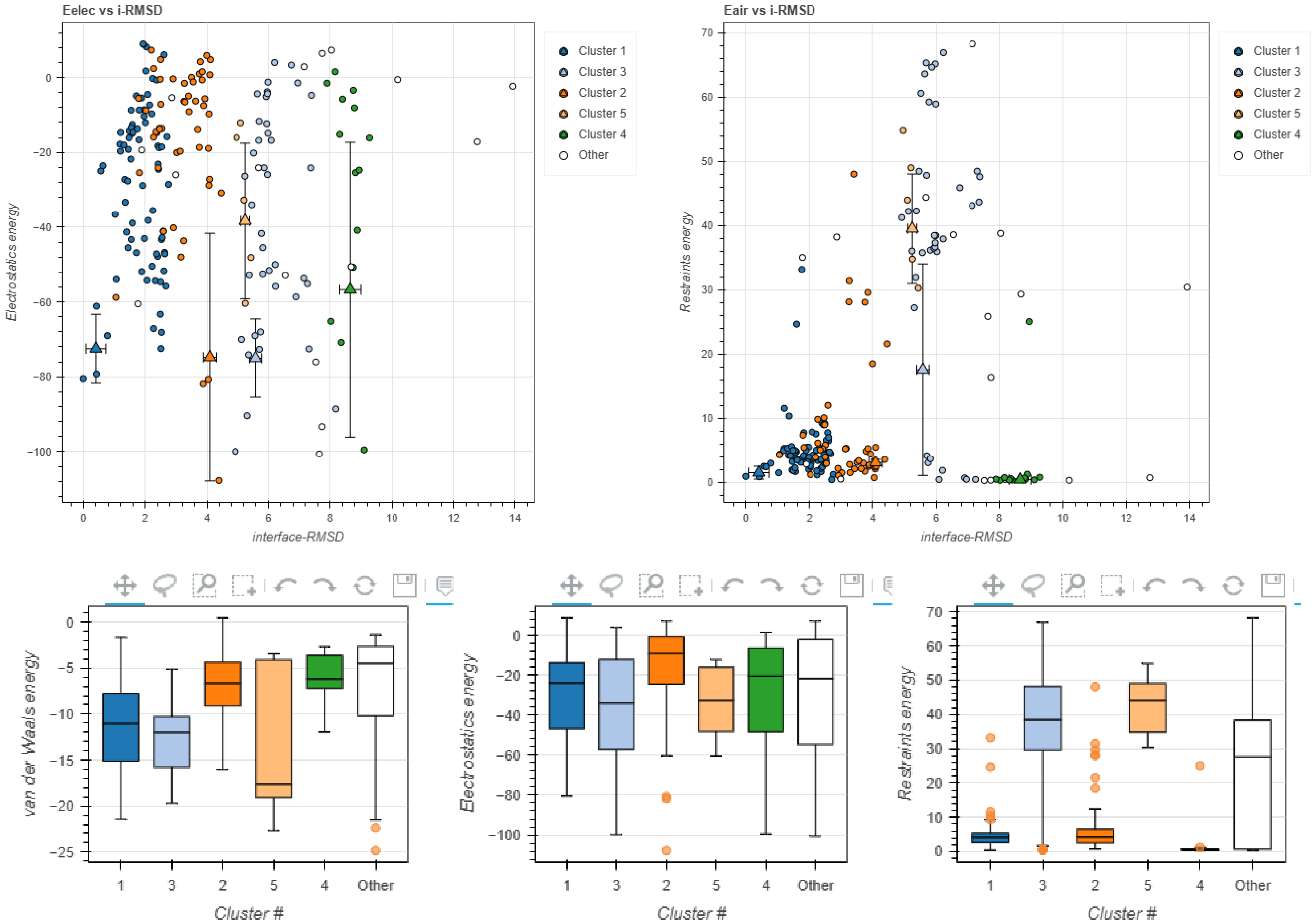
HADDOCK Score	-40.0 +- 2.0
Van der Waals Energy	-20.2 +- 0.7
Electrostatic Energy	-72.5 +- 7.9
Desolvation Energy	-5.5 +- 1.0
Buried Surface Area	540.2 +- 10.1 Å²

Electrostatic dominance: largest contribution to the stability of -72.5 kcal/mol is electrostatic interaction, inhibitor successfully mimics the salt bridge of Asp366

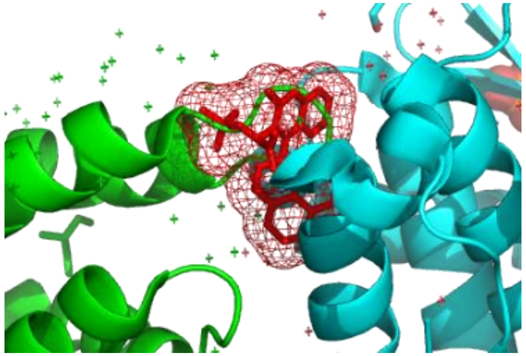
Hydrophobic complementarity: Van der Waals energy (-20.2 kcal/mol) indicates that aromatic skeleton of BI-224436 perfectly packs into the orange hydrophobic cleft, displacing water molecules from there (positive desolvation).

Pose stability: low RMSD (0.2Å) inside cluster indicates that inhibitor has a single rigidly fixed conformation in the pocket


HADDOCK score vs FCC
HADDOCK score
Fraction of Common Contacts
0
-10
-20
-30
-40
0
0.2
0.4
0.6
0.8
1
Cluster 1
Cluster 3
Cluster 2
Cluster 5
Cluster 4
Other
HADDOCK score vs i-RMSD
HADDOCK score
interface-RMSD
0
-10
-20
-30
-40
0
2
4
6
8
10
12
14
Cluster 1
Cluster 3
Cluster 2
Cluster 5
Cluster 4
Other
Edesolv vs i-RMSD
Desolvation energy
interface-RMSD
2
0
-2
-4
-6
-8
-10
0
2
4
6
8
10
12
14
Cluster 1
Cluster 3
Cluster 2
Cluster 5
Cluster 4
Other
Evdw vs i-RMSD
van der Waals energy
interface-RMSD
0
-5
-10
-15
-20
-25
0
2
4
6
8
10
12
14
Cluster 1
Cluster 3
Cluster 2
Cluster 5
Cluster 4
Other

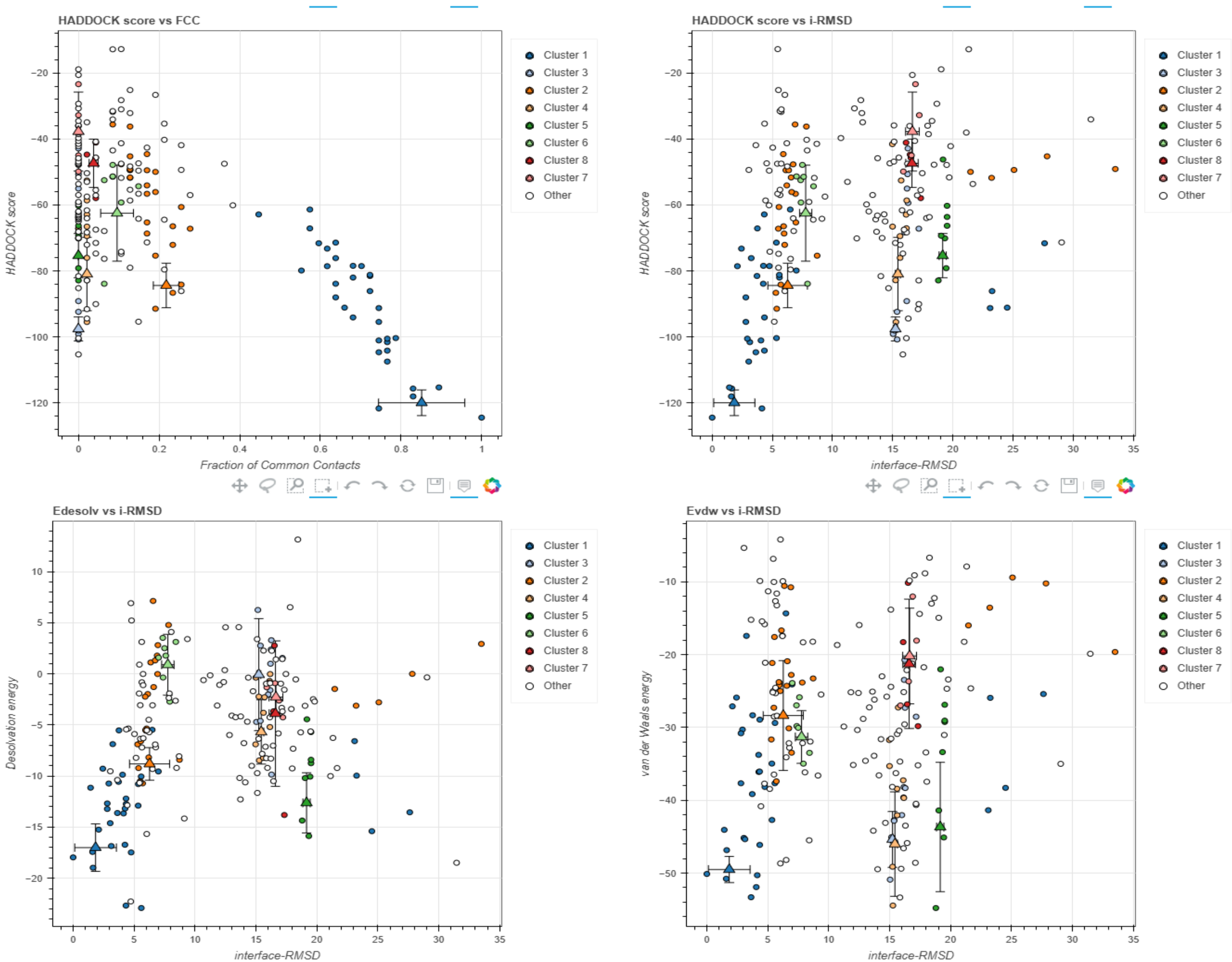


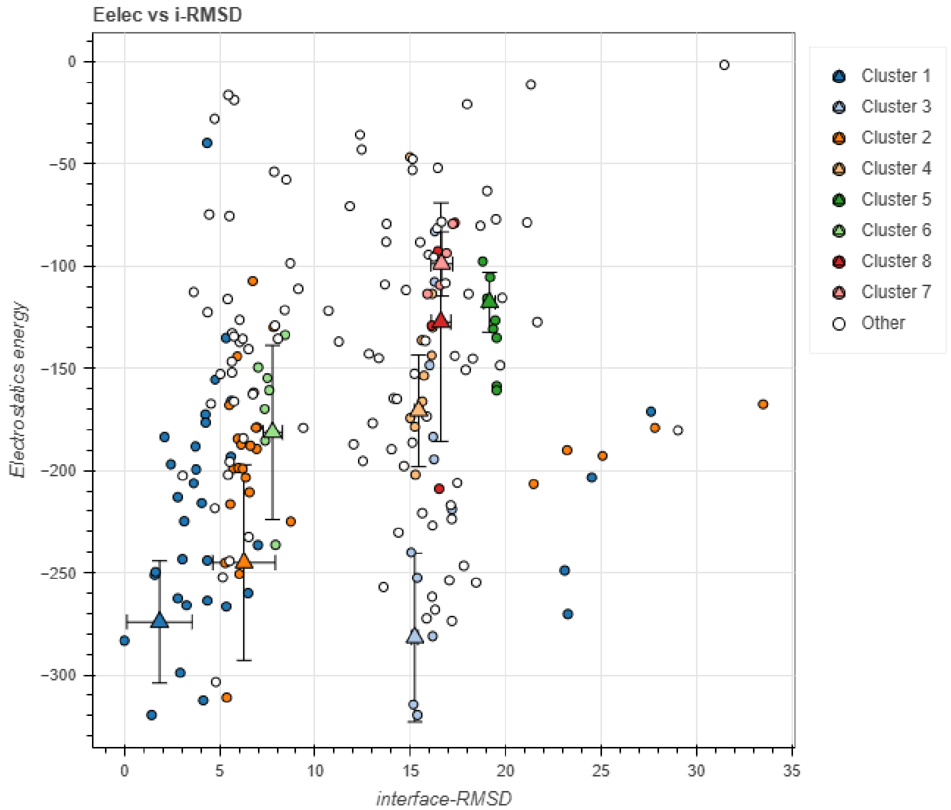
Eelec vs i-RMSD
Electrostatics energy
interface-RMSD
Eair vs i-RMSD
Restraints energy
interface-RMSD
Cluster 1
Cluster 3
Cluster 2
Cluster 5
Cluster 4
Other
van der Waals energy
Cluster #
Electrostatics energy
Cluster #
Restraints energy
Cluster #

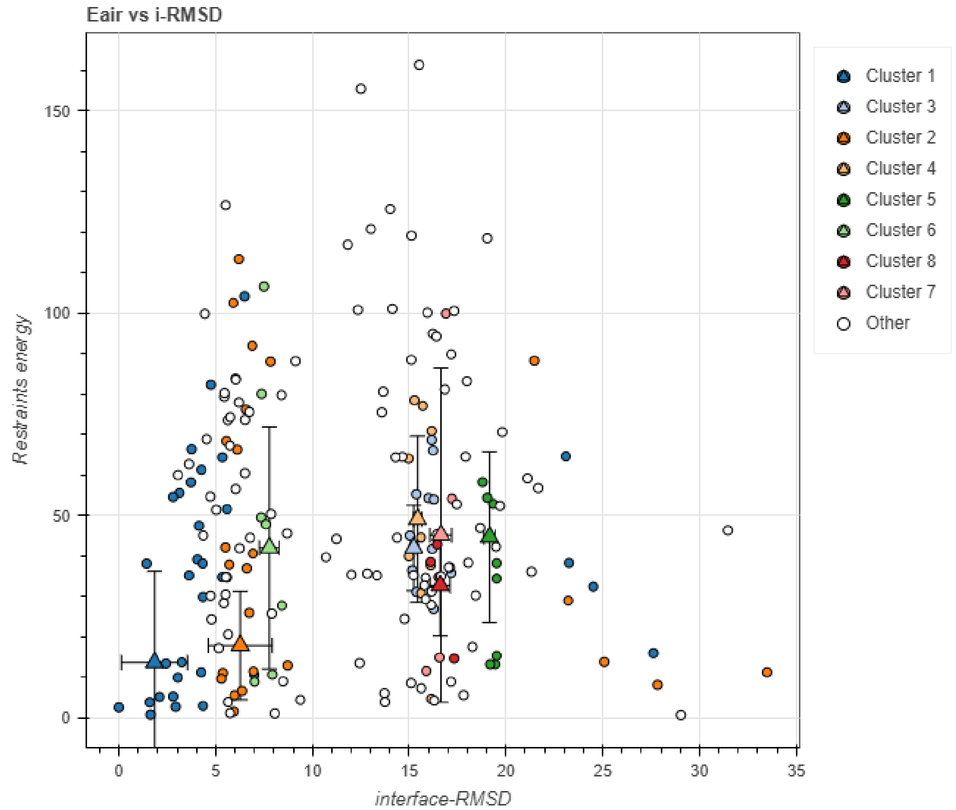


Part IV: Mechanism of competitive inhibition (3-Body HADDOCK & Steric Clash)

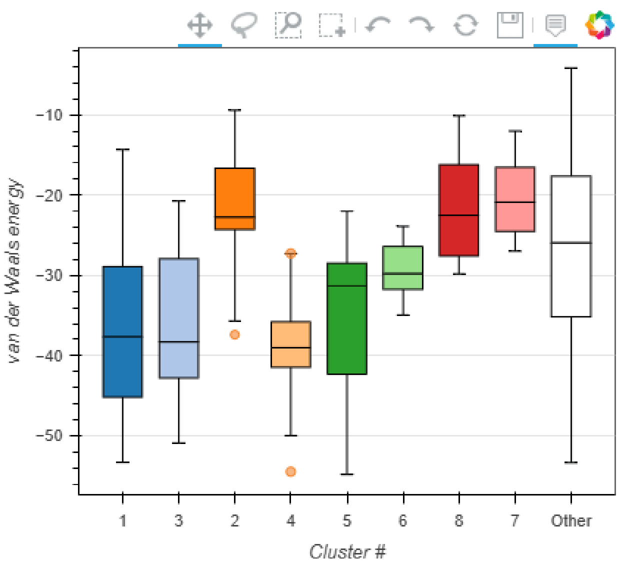
To confirm inhibition mechanism, there was performed multibody docking (integrase dimer + LEDGF cofactor + BI-224436 inhibitor). top cluster 1 (HADDOCK Score -120.0, Z-Score -1.8) show anomalously high constraint violation energy (Restraints violation energy = 13.6 ± 19.6 kcal/mol), which indicates physical impossibility of simultaneous binding of both ligands, and prove that ALLINIs act as direct disruptors of protein-protein interactions, forming a rigid steric block (clash) for the cofactor loop.

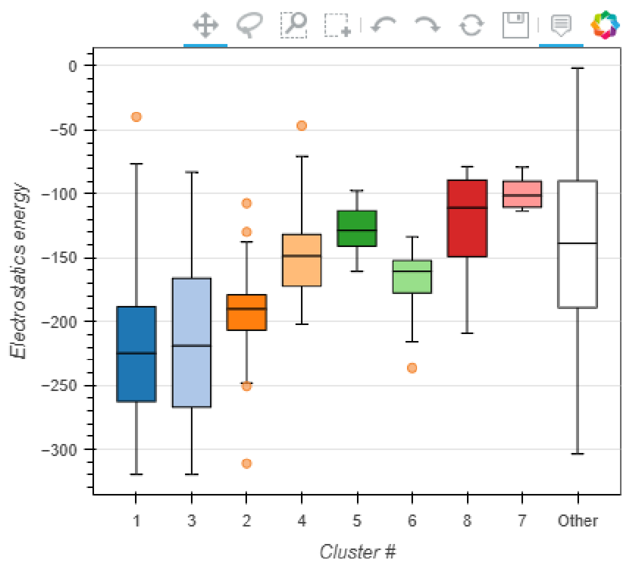


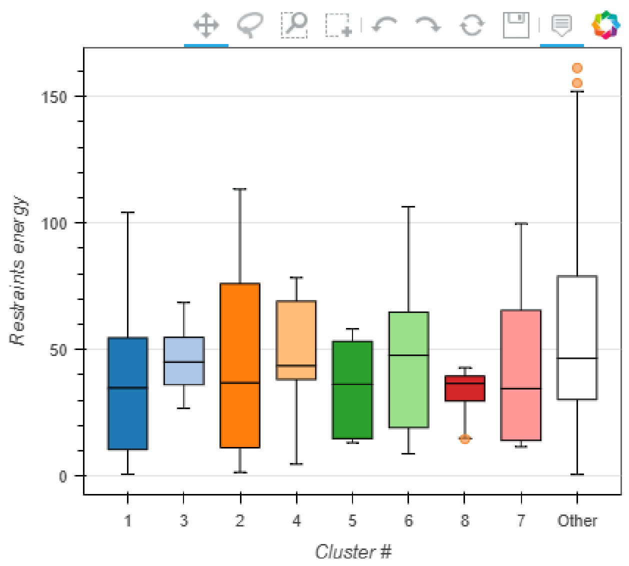




Cluster Analysis







MODULE 2: Protein Binder Design and Optimization

Part V: Identification and validation of the protein scaffold (1QX3)

after confirmation of validity of allosteric pocket using small inhibitor, next step was to search a protein disruptor -- scaffold -- capable of spatially filling the V-shaped cleft of integrase and reproducing inhibition effect.

5.1. HECTOR Inputs: due to topology of the target pocket, scan was through the key atoms of interchain interface.

Parameters of most successful simulation, which allowed finding highly complementary structure:

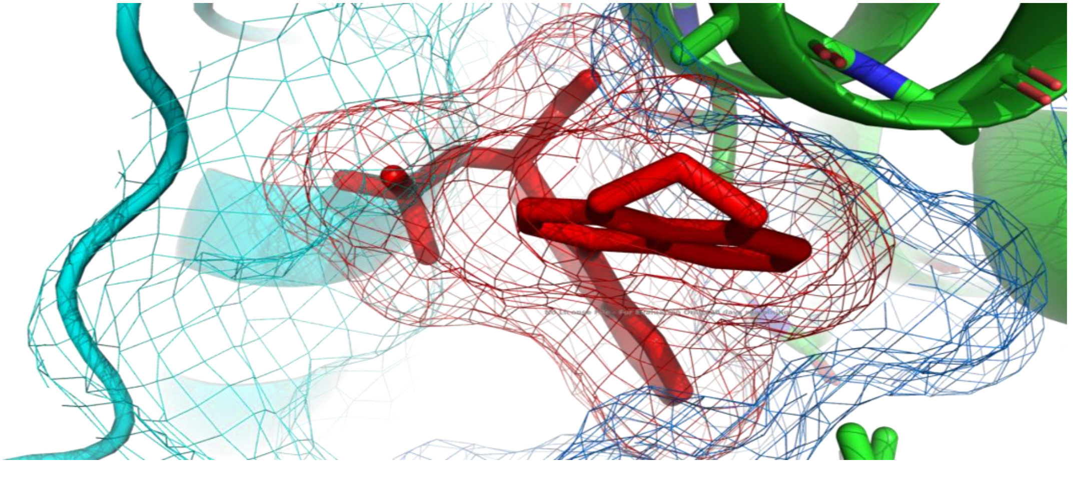
- **Target Structure:** HIV-1 integrase dimer.
- **Anchors:** residue 247 (Atom: CB) or **Gln148**, residue 70 (Atom: CB) or **Thr125**.
- **R-factor threshold:** -0.5
- **RMSD threshold:** 4.0 Å
- **Minimum amount of Interface Residues:** 10
- **Clash threshold:** 100

5.2. Optimization Process: Various scan vectors and stiffness thresholds were tested, but previous attempts with other vectors had significant steric clash threshold (to Clash = 500). Changing the scan axis to a vector (247 CB — 70 CB) was a good idea. algorithm filtered out irrelevant structures and found single scaffold that fit perfectly into the cleft without critical violations of dimer geometry.

Given topology of the target pocket and technical end-to-end numbering of nput PDB (offset of -55 residues relative to UniProt), scan vector was laid through the key atoms of interchain interface:

- Anchor 1: Residue 70 (Atom: CB) — spatially corresponds to Thr125 on the first monomer (Chain A).
- Anchor 2: Residue 247 (Atom: CB) — spatially corresponds to Gln148 on the second monomer (Chain B).

5.3. Analysis of results (1QX3): absolute leader of the screening was **1QX3**.



Average energy per amino acid during minimization:

- t = 0%: 6070.28 kcal/mol
- t = 10%: -13.69 kcal/mol
- t = 20%: -17.9 kcal/mol
- t = 30%: -18.65 kcal/mol
- t = 40%: -19.03 kcal/mol
- t = 50%: -19.22 kcal/mol
- t = 60%: -19.46 kcal/mol
- t = 70%: -19.59 kcal/mol
- t = 80%: -19.69 kcal/mol
- t = 90%: -19.75 kcal/mol
- t = 100%: -19.82 kcal/mol

overlap of finded binder 1QX3 (cyan) and small inhibitor (red) in integrase allosteric pocket.

5.4. OpenMM: Due to high initial complementarity of the 1QX3, algorithm was stabilized the system extremely quickly: potential energy of the complex dropped rapidly from the initial 6070.28 kcal/mol to a stable -19.82 kcal/mol. This proves successful and thermodynamically favorable placement of the 1QX3 binder in allosteric pocket. This structure can be a potential a basic template for further in silico mutagenesis.

Conclusions

1. **Target validation and mechanism of inhibition:** study confirmed the thermodynamic stability of the HIV-1 integrase homodimer (Delta G = -12.3 kcal/mol) and validated the V-shaped allosteric cleft as a target.
2. **Binder design:** 1QX3 protein scaffold was identified by geometric screening (HECTOR algorithm) using a highly specific scanning vector (Thr125–Gln148). multibody docking proven that small molecules of the ALLINIs class (BI-224436) act through direct steric competition, physically blocking the access of the LEDGF/p75 cofactor
3. **Energetic stabilization:** relaxation of the Integrase–1QX3 complex (OpenMM/Damietta) allowed to remove the steric clash and achieve a stable energy minimum (-19.82kcal/mol). absence of critical steric conflicts after minimization confirms the high complementarity
4. **Perspectives:** WT 1QX3 protein can be lead candidate for further development of protein-based integrase inhibitors. Next step is to optimize its affinity by rational in silico mutagenesis of side chains to maximize hydrophobic and electrostatic interactions in the depth of the cleft.

NB. Due to an unforeseen technical failure of laptop and the loss all the project info, this report contains only those results and simulations that were successfully restored.

https://github.com/inna-tsymb/structural_bioinformatics/tree/main/homework3-4/